

JOBSAGE — Developer Handover Pack

Decision Intelligence Platform for Regulated Professional Mobility (UK)

Document purpose: handover for software development (MVP scope).

Source: "Product Description and a Preliminary Business Plan" (publicly available regulatory guidance is encoded; no regulator database access; user-declared data only).

1. Product Overview

What we are building: A pre-application decision layer that determines which regulated UK roles a professional is **legally eligible** to pursue now, and what steps will make them eligible in future. This is not a job board and not a recruitment agency workflow.

Core problem: Existing job boards and staffing workflows list roles without assessing regulatory eligibility, licence readiness, or visa sponsorship feasibility, generating high volumes of ineligible applications and delaying hiring in shortage sectors.

Design principle: Eligibility before opportunity. Any role shown to a candidate must satisfy (1) regulatory eligibility, (2) licence-to-practise readiness where applicable, and (3) sponsorship feasibility if required.

Target users

- **Candidates:** regulated healthcare professionals and academic/clinical academic candidates seeking UK roles (UK-trained, overseas-trained, UK-resident internationals).
- **Institutions (phase 1b+):** NHS Trusts and UK universities needing compliance-filtered pipelines and reduced screening burden.
- **Reviewers/Admin:** internal staff managing rule updates, audits, and edge-case review.

2. MVP Scope (Build First)

2.1 Candidate Journey (MVP)

- 1 Create account and provide consent.
- 2 Complete structured profile (qualifications, experience, registration status, residency/sponsorship need).
- 3 Run eligibility evaluation via rules engine.
- 4 If eligible: show viable roles only (filtered by eligibility + sponsorship).
- 5 If not yet eligible: show "Not Yet Eligible" remediation plan (gaps, steps, indicative timelines/cost ranges).
- 6 Optional: request human review for edge cases.

2.2 Out of Scope (MVP)

- No regulator database access and no registration verification. Use publicly available guidance only.
- No guarantee of visa sponsorship; feasibility modelling only with disclaimers.
- No end-to-end job application submission required for MVP (platform is a pre-application decision layer).

3. Functional Requirements

FR1 Authentication & Consent

- Email/password authentication (SSO optional later).
- Consent capture and storage (timestamp, terms version).

FR2 Candidate Profile (Structured, Declarative)

- Profession and specialty (where relevant).
- Country of qualification; qualification type/year.
- Experience years and role history summary.
- Declared registration status and licence readiness fields.
- Residency/visa status and “requires sponsorship” boolean.
- Optional document uploads (stored securely; not verified).

FR3 Regulatory Rules Engine (Core IP)

- Encode public guidance for initial regulators: GMC, NMC, HCPC.
- Evaluate user-declared inputs → outcomes: Eligible now / Not yet eligible / Ineligible.
- Return structured reason codes and user-facing explanation text.
- Version-controlled rulesets with effective dates and changelog.
- Deterministic and auditable outputs (trace to rule IDs and ruleset version).

FR4 Visa Sponsorship Feasibility

- Assess sponsorship feasibility for roles when user requires sponsorship.
- Output: feasible / not feasible / uncertain + reason codes.
- Always show disclaimers: no guarantee of sponsorship.

FR5 Opportunity Matching & Filtering

- Maintain role catalogue (CSV import for MVP; API later).
- Filter roles by eligibility + licence + sponsorship feasibility.
- Expose “why this role is shown” explanations.

FR6 “Not Yet Eligible” Remediation Journey

- Explain unmet eligibility and list specific gaps.
- Generate ordered action plan steps with indicative timeline and cost ranges.
- Allow user to track progress (planned / in progress / done).

FR7 AI-Assisted Decision Intelligence (Bounded)

- AI may prioritise pathways, estimate timelines, flag risks, and suggest sequencing.
- AI must never override rules engine eligibility outcomes.
- UI must label AI outputs as recommendations.

FR8 Human-in-the-Loop Review

- System flags ambiguous/high-risk cases to reviewer queue.
- Reviewer can annotate guidance and recommended pathways.
- All reviewer actions are audited.

FR9 Compliance & Audit

- Decision traceability: inputs snapshot hash, ruleset version, outcome, reason codes.
- Admin audit logs for rule changes and manual overrides.
- Export of audit logs/decisions (CSV/JSON).

4. NonFunctional Requirements

Security & Privacy

- Encrypt data in transit and at rest.
- Role-based access control (candidate/admin/reviewer/institution).
- Minimise PII and enforce least-privilege access.

Explainability & Trust

- Every eligibility result must include a plain-language explanation and next steps.
- Prominent disclaimers: platform is decision-support; regulators make final decisions.
- Full traceability of which ruleset produced the result.

Maintainability

- Rules engine is modular; updates should not require rewriting core application logic.
- Regression tests for rule updates with sample profiles and expected outcomes.

Scalability

- Add new professions and jurisdictions by adding modular rulesets and mappings.
- Support increased user volume with stateless services and scalable storage.

5. MVP Backlog — Epics, User Stories & Acceptance Criteria

Epic A — Authentication, Consent & Account

A1 — Register account

As a candidate, I want to create an account so I can save my profile and eligibility results.

Acceptance Criteria:

- User can register using email + password.
- System validates email format and password strength.
- Consent to terms/privacy is stored with timestamp and terms version.
- User is redirected to profile onboarding after registration.

A2 — Login/Logout

As a user, I want to log in/out securely.

Acceptance Criteria:

- Login works with valid credentials.
- Invalid login returns generic error (no user enumeration).
- Logout invalidates session/token.

Epic B — Candidate Profile

B1 — Create/edit structured profile

As a candidate, I want to enter structured qualification and status information.

Acceptance Criteria:

- Required fields are captured (profession, qualification country, experience, registration status, residency/sponsorship need).
- Validation prompts user for missing mandatory fields.
- Profile is editable and versioned for decision trace purposes.

B2 — Optional document uploads

As a candidate, I want to upload documents optionally.

Acceptance Criteria:

- Supported types: PDF/JPG/PNG (configurable); max size enforced.
- Files stored in secure object storage.
- UI shows disclaimer: uploads are not verified and do not constitute approval.

Epic C — Regulatory Rules Engine

C1 — Evaluate eligibility

As a candidate, I want the system to determine my likely eligibility status based on regulatory pathways.

Acceptance Criteria:

- Rules engine returns Eligible now / Not yet eligible / Ineligible.
- Output includes reason codes + user-friendly explanation.
- Decision record stores ruleset version used and input snapshot hash.
- UI displays disclaimers that final decisions rest with regulators.

C2 — Show applicable pathways and gaps

As a candidate, I want to see relevant routes and missing steps.

Acceptance Criteria:

- System lists applicable pathways (e.g., route names) and per-pathway prerequisite status.
- Each pathway includes missing steps and indicative timeline estimate (when available).
- Pathway details reference rule IDs and ruleset version.

C3 — Admin rule management

As an admin, I want to update rules safely with versioning.

Acceptance Criteria:

- Admin can draft a new ruleset version and maintain changelog.
- Publishing a ruleset requires admin permission.
- Existing decision records remain linked to original ruleset version.
- Regression tests can be executed against sample profiles.

Epic D — Sponsorship Feasibility

D1 — Sponsorship feasibility per role

As a candidate who requires sponsorship, I want to see if roles are feasibly sponsorable.

Acceptance Criteria:

- User can indicate “requires sponsorship”.
- For each role: feasible / not feasible / uncertain returned with reason codes.
- Disclaimers state this is not a guarantee of sponsorship.

Epic E — Opportunity Matching

E1 — View viable roles only

As a candidate, I want to see only roles I am eligible to pursue.

Acceptance Criteria:

- Roles are filtered by regulatory eligibility + licence readiness + sponsorship feasibility (if required).
- Ineligible roles are excluded from candidate view.

- Each shown role includes an explanation of why it is shown.

E2 — Admin role import

As an admin, I want to upload/import job roles into the platform.

Acceptance Criteria:

- Admin can upload CSV using provided template.
- System validates required columns and shows per-row errors.
- Imported roles appear in catalogue and become matchable.

Epic F — Not Yet Eligible Journey

F1 — Show remediation plan

As a candidate who is not eligible, I want a structured plan showing what to do next.

Acceptance Criteria:

- If outcome is Not yet eligible, system shows gaps + ordered remediation steps.
- Plan includes indicative timeline and cost range fields (configurable).
- User can mark steps planned/in progress/done; progress persists.

F2 — Bounded AI prioritisation

As a candidate, I want suggested prioritisation of steps based on time and readiness.

Acceptance Criteria:

- AI suggestions appear only as recommendations within remediation plan.
- AI does not change eligibility outcome.
- User can override AI ordering; system stores final ordering.

Epic G — Human Review

G1 — Flag edge cases

As the system, I want to flag profiles requiring human review.

Acceptance Criteria:

- Rules engine or missing data triggers a review flag.
- Candidate is informed and given expected review timeline.
- Case appears in reviewer queue.

G2 — Reviewer workflow

As a reviewer, I want to review flagged cases and leave guidance.

Acceptance Criteria:

- Reviewer can view profile snapshot and rule evaluation output.
- Reviewer can add notes and recommended pathway guidance.
- All reviewer actions are captured in audit log.

Epic H — Compliance & Audit

H1 — Explainability & disclaimers

As a user, I want clear explanations and disclaimers on all outcomes.

Acceptance Criteria:

- Every decision view shows explanation + next steps + disclaimers.
- Consent logs are recorded and retrievable by admin.
- Decision record includes ruleset version and reason codes.

H2 — Export decision traces

As an admin, I want to export decision logs for audit.

Acceptance Criteria:

- Admin can export audit logs/decisions in CSV/JSON.
- Exports include anonymised user ID, timestamp, ruleset version, outcome, and reason codes.

6. Technical Architecture (Proposed)

Architecture pattern: API■first modular services with a deterministic rules engine as the source of truth, a bounded AI recommendation service, and human-in-the-loop review. All decisions are auditable and traceable to ruleset versions.

6.1 Logical Diagram (text)

```
[CANDIDATE UI] ----> [API/BACKEND] ----> [DECISION ORCHESTRATOR]
|                                     |----> [RULES ENGINE (versioned)] ---> [RULES STORE]
|                                     |----> [VISA/SPONSORSHIP SERVICE]
|                                     |----> [AI RECOMMENDER (bounded)] ---> [FEATURE STORE]
|                                     |----> [HUMAN REVIEW SERVICE] ---> [REVIEW QUEUE]
|----> [PROFILE SERVICE] ---> [PROFILE DB]
|----> [JOBS CATALOG SERVICE] ---> [ROLES DB]
|----> [AUDIT/TRACE SERVICE] ---> [AUDIT LOG STORE]
|----> [DOC UPLOAD SERVICE] ---> [OBJECT STORAGE]
```

[ADMIN CONSOLE] and [REVIEWER CONSOLE] connect to the API and respective services.

6.2 Minimum Data Entities

- **CandidateProfile:** structured declarative inputs (profession, qualification country, experience, registration status, residency, requires_sponsorship, etc.).
- **Ruleset/Rule:** regulator, pathway, conditions DSL, outcomes, reason codes, version, effective dates, changelog.
- **DecisionRecord:** user reference, input snapshot hash, ruleset version used, outcome, reason codes, explanation, timestamp.
- **RemediationPlan:** gaps + ordered steps with status tracking, indicative timelines and cost ranges.
- **JobRole:** role metadata + requirements (registration/licence/sponsorship flags) to enable filtering.

6.3 Rule Engine Implementation Guidance

- Use a declarative DSL (JSON/YAML) for rules to simplify updates and auditing.
- Each rule evaluation returns reason codes and a trace path (which rules fired).
- Maintain regression tests using curated sample profiles and expected outcomes.
- Keep AI outputs separate and clearly marked as recommendations.

7. Suggested Engineering Delivery Plan

- Sprint 0–1: Auth, consent, profile scaffolding, audit logging foundation.
- Sprint 2–3: Rules engine v1 (pilot regulator/pathway), decision records, explanation UI.
- Sprint 4: Not Yet Eligible plan generator + progress tracking.
- Sprint 5: Role ingestion (CSV), matching/filtering, role detail pages.
- Sprint 6: Human review queue + reviewer console, admin exports.
- Sprint 7: Bounded AI recommendations (timeline/prioritisation/risk flags) with guardrails.

Appendix: Mandatory Disclaimers (UI Copy Seed)

- This platform provides decision-support guidance based on publicly available information and user-declared inputs.
- We do not verify registrations or access regulator databases. Final decisions remain with the relevant UK regulator and/or employer.
- Visa and sponsorship feasibility indicators are informational only and do not guarantee sponsorship or visa outcomes.